









Statistics of Financial Markets

Chapter 0: Introduction



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Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ASE**: Academia de Studii Economice din București (the Bucharest University of Economic Studies)
- **BET**: Bucharest Exchange Trading (index)
- **BET-TR**: BET Total Return (index)
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **COVID-19**: COronaVIrus Disease 2019
- **CSIE**: Facultatea de Cibernetică, Statistică și Informatică Economică (the Faculty of Cybernetics, Statistics and Economic Informatics)
- **EOD**: End Of Day (daily closing data)
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **ETF**: Exchange-Traded Fund
- **EU**: European Union

Acronyms used in this material (2/2)

- **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- **FX**: Foreign eXchange
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **IPO**: Initial Public Offering
- **LLM**: Large Language Model
- **MDD**: Maximum DrawDown
- **NYSE**: New York Stock Exchange
- **SEC**: Securities and Exchange Commission
- **SFE**: Statistics of Financial Markets, Exercises (Quantlet collection of the textbook)
- **US**: United States
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- **VIX**: CBOE Volatility IndeX
- **VOC**: Vereenigde Oostindische Compagnie (the Dutch East India Company)
- **WIG20**: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Route of This Chapter

- **Guiding questions**
  - what does a statistician measure on a market?
  - which numbers describe 2026 markets?
  - how do we get from prices to returns?
- **Part I**
  - how the course works: evaluation, textbook, chapters, AI policy
  - why financial markets matter, and what the 2026 data show
- **Part II**
  - a short history of exchanges, crashes and models
  - returns and first statistics, a teaser
  - AI for scientific discovery: one open question
- **After this chapter you can**
  - explain how the course is organised and graded
  - name the main asset classes and market participants
  - read a growth chart, a risk–return chart and a drawdown chart
  - compute a simple and a log return from two prices
  - state what must be checked in an AI-generated analysis
- Seminar 0 comes **before** this lecture and practises the same tools on data

What Statistics of Financial Markets Studies

- **Statistics of financial markets**: statistical methods for prices, returns and risks of traded assets
  - describe: how large and how variable are returns?
  - model: which distribution, which dependence over time?
  - measure risk: how large can a loss be on a bad day?
- Why financial data need their own methods
  - extreme days are far more frequent than the Normal distribution predicts (*heavy tails*)
  - calm and turbulent periods alternate (*volatility clustering*)
  - markets fall together in crises: diversification weakens when it is most needed
- These regularities are the *stylised facts* of returns (Cont (2001)), studied in Chapter 2

Course Overview

- **Programme**
  - Bachelor's programme Statistics and Data Science, year 3, semester 2
  - Faculty of Cybernetics, Statistics and Economic Informatics (CSIE), ASE Bucharest
  - academic year 2026/2027
- **Lecture**
  - Prof. Daniel Traian Pele, danpele@ase.ro
- **Prerequisites**
  - probability and statistics, linear algebra, Python
- **Course website**
  - danpele.github.io/SFM: slides, seminars, notebooks, Quantlets, quizzes



The Bucharest University of Economic Studies (ASE)

Photo: Joe Mabel (2014); CC BY 3.0; Wikimedia Commons

Evaluation

- **Written exam: 70%**
  - problems on real data and short interpretation questions, as in the seminars
  - covers the lectures and the seminars of Chapters 0–16
- **Team project: 20%**
  - one research question on real market data, answered with the methods of the course
- **Attendance and quizzes: 10%**
  - lectures and seminars
  - the chapter quizzes on the website, taken with the ASE account; name and score are recorded

Textbook and Further Reading

- **Main textbook**
  - Franke, Härdle & Hafner (2019), *Statistics of Financial Markets: An Introduction*, 5th edition, Springer
  - the chapters of the course follow its parts on returns, distributions, time series, volatility and risk
  - its code is public as the SFE Quantlets
- **For the seminars**
  - Borak, Härdle & López-Cabrera (2013), *Statistics of Financial Markets: Exercises and Solutions*, Springer
- **Further reading**
  - Tsay (2010), *Analysis of Financial Time Series*
  - Campbell, Lo & MacKinlay (1997), *The Econometrics of Financial Markets*
  - the Statistics of Financial Markets course on Quantinar, a peer-to-peer learning platform

Course Map: Chapters 0–16

No.	Chapter
0	Introduction
1	Data, returns and indicators
2	Classical distributions and stylised facts
3	α-stable distributions
4	Probability
5	Heavy tails and extreme value theory
6	Model selection and risk management
7	Efficient markets, random walk and variance-ratio tests
8	Volatility estimators

No.	Chapter
9	ARCH and GARCH models
10	VaR, ES and backtesting
11	Fractal markets hypothesis and long memory
12	Scoring models
13	Machine learning
14	Crypto assets
15	Systemic risk
16	Review

- Data and distributions (1–6), market efficiency (7), volatility and risk (8–11), applications (12–15), review (16)

Materials and Tools

- **For every chapter**
  - lecture slides and seminar slides, in English and in Romanian
  - a lecture notebook and a seminar notebook (Python), opened in Google Colab with one click
  - a self-assessment quiz on the website: 20 questions drawn from a bank of 24, each answer explained
- **Quantlets**: every chart has a public folder with code, a description file (`Metainfo.txt`) and the chart
  - the icon under each chart opens its Quantlet on GitHub
- **Data**: daily market data from EODHD (EOD Historical Data), saved once in the course repository
  - the same numbers on every computer, until 18 September 2026
  - EUR/RON: the official reference rate of the BNR (National Bank of Romania)

Seminars

- **Each seminar comes before its lecture**
  - it starts with a short primer, so it can be followed without the lecture
  - the lecture then explains why the tools work and where they fail
- **Two kinds of exercises**
  - **[Solved]**: full solution in the slides and in the notebook, a model to copy
  - **[Proposed]**: you solve it; the solution is discussed in class
- **Nothing is handed in**
  - homework is practice for the exam and for the project
  - one exercise in each seminar asks you to find the error in an AI-generated answer

The Team Project

- **Question**: one research question about real market data, chosen by the team
- **Deliverables**
  - a GitHub repository whose code reproduces every number and every chart from the saved data
  - a short report and a presentation of the results
  - the file `AI_USE.md` described in the AI policy (next slide)
- **What we grade**
  - a clear question, correct methods and checks, an honest interpretation
  - **oral defence**: each member explains the code and the results
- Project seeds appear at the end of every chapter, in the section *AI for scientific discovery*

AI Policy

- **AI tools are allowed and must be declared**
  - AI: artificial intelligence; here, assistants based on an LLM (large language model) that write text and code
  - every use goes into `AI_USE.md`: the tool, the request, what was kept, what was corrected
- **You are responsible for every number and every reference**
  - typical AI errors: invented references, wrong formulas, code that runs but computes the wrong thing
  - every reference: open its DOI (Digital Object Identifier) and check the title
  - every number: recompute it with your own code from the course data
- **Oral defence**
  - if you cannot explain a line of your code, it does not count as your work

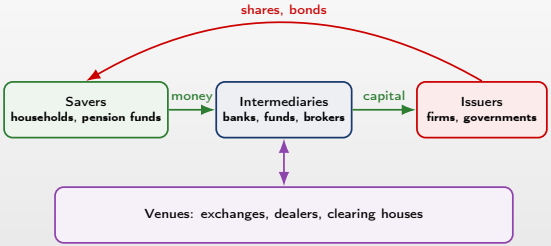
Recap: Organisation

- Grade: 70% written exam, 20% team project, 10% attendance and quizzes
- Textbook: Franke, Härdle & Hafner (2019); seminars: Borak, Härdle & López-Cabrera (2013)
- Seminars come before lectures; nothing is handed in
- AI is allowed, declared in `AI_USE.md`, and checked at the oral defence

What a Financial Market Does

- **Financial market**: a place or a system where buyers and sellers exchange claims on future cash flows (shares, bonds, currencies, contracts, tokens) at prices set by supply and demand
- **Five functions**
  - **capital allocation**: savings reach firms and governments
  - **risk sharing**: risks move to those willing to carry them
  - **liquidity**: an asset can be turned into cash quickly
  - **price discovery**: prices gather information spread across many traders
  - **payments and settlement**: trades are completed safely
- For a statistician, the market is a data generator: every trade leaves a price and a time

Who Trades



- **Buy side**: investors who buy assets: individuals (retail), pension funds, insurers, asset managers
- **Passive** or **active** investing
  - passive: copy an index, for example through an ETF (exchange-traded fund, a fund traded like a share)
  - active: try to beat the index
- **Sell side**: brokers execute orders; market makers quote a buy and a sell price all day

Asset Classes

Class	Examples	Cash flow	Main risks
Equities	shares, equity ETFs	dividends	market, firm
Bonds	government and corporate bonds	coupons, principal	interest rate, credit
Currencies (FX)	EUR/USD, EUR/RON	interest differential	currency, policy
Commodities	gold, oil	none	supply, storage
Derivatives	futures, options	contingent payoffs	leverage, counterparty
Crypto assets	Bitcoin, Ethereum	none	volatility, regulation

- FX (foreign exchange): the currency market; EUR/RON = lei paid for one euro
- One statistical toolbox for all of them: returns, volatility, tails and dependence

How a Price Is Formed: the Order Book



- **Order book**: all waiting buy orders (**bids**) and sell orders (**asks**), sorted by price; bars show quantities
- **Bid–ask spread**: best ask minus best bid, here $100.01 - 99.99 = 0.02$, the cost of trading at once
- A **market order** buys or sells at once at the best prices available
- **Questions for the room**: a market order to buy 5 units arrives
  - at what average price is it filled?
  - what is the spread afterwards?

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  - what is the spread afterwards?
- **Answer**
  - 2 units at 100.01 and 3 at 100.02: average $(2 \times 100.01 + 3 \times 100.02)/5 = 100.016$
  - the best ask becomes 100.02, the spread widens to 0.03: the order moved the price

The Role of Statistics

- **Tomorrow's price is uncertain**: we model it as a random variable
  - P_t: the price on day t; the sequence $P_1, P_2, \ldots$ is a **time series**
- **What we measure in this course**
  - **return**: the relative change of the price (Chapter 1)
  - **distribution** of returns, and how heavy its tails are (Chapters 2, 3 and 5)
  - **volatility**: the standard deviation of returns, and how it changes in time (Chapters 8 and 9)
  - **risk measures**: VaR (Value at Risk) and ES (Expected Shortfall) (Chapter 10)
- **The big question**: can past prices predict future returns?
  - the efficient market hypothesis says: hardly (Chapter 7)

Recap: the role of financial markets

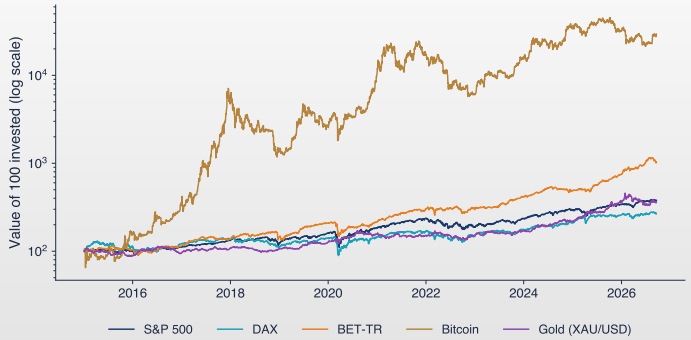
- Markets allocate capital, share risk, provide liquidity and discover prices
- Prices come from the order book: the spread is the cost of trading at once
- Six asset classes, one statistical toolbox
- Tomorrow's price is a random variable: statistics describes its returns, volatility and risk

Data Used in This Course

Series	What it measures	Price	Calendar
S&P 500	500 large US firms	close	US trading days
DAX	40 large German firms	close	German trading days
BET, BET-TR	largest firms of the BVB (Bucharest Stock Exchange)	close	Romanian trading days
Gold	XAU/USD, dollars per ounce	close	weekdays
Bitcoin	BTC/USD	close at 24:00 UTC	every day
VIX	expected S&P 500 volatility	close	US trading days

- Daily closes until 18 September 2026; source: Materials and Tools
- BET-TR (total return): BET with dividends reinvested; UTC: Coordinated Universal Time
- Each series keeps its own calendar: Bitcoin trades 365 days a year, exchanges about 252

Five Markets since 2015: Growth of 100



- Value of 100 invested in January 2015, log scale: equal vertical distances are equal percentage changes
- Indices in local currency, gold and Bitcoin in USD

SFM_ch0_markets

Five Markets since 2015: the Numbers

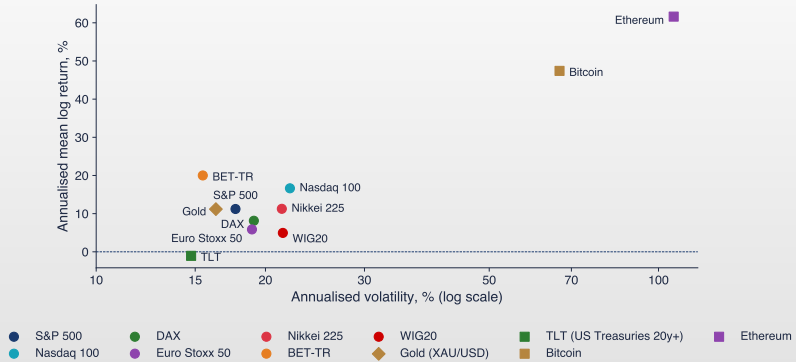
	P_T/P_0	Return	Volatility
S&P 500	$\times 3.7$	11.2	17.7
DAX	$\times 2.6$	8.1	19.1
BET-TR	$\times 10.4$	20.0	15.5
Gold	$\times 3.7$	11.2	16.3
Bitcoin	$\times 257.4$	47.4	66.7

- P_T/P_0: final price over initial price; return and volatility in % a year

- **Return p.a.**: average daily log return times A, the number of trading days per year
- **Volatility p.a.**: standard deviation of daily returns times $\sqrt{A}$ (definitions in Part II)
- **Reading**
  - Bitcoin: the highest return and about four times the volatility of the stock indices
  - BET-TR beat the S&P 500 with lower volatility over this window
  - one window of 11 years: a different start date gives a different ranking

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Risk and Return, 2015–2026



⊡ Each point: one market; horizontal axis volatility (log scale), vertical axis mean return, both per year

SFM_ch0_risk_return

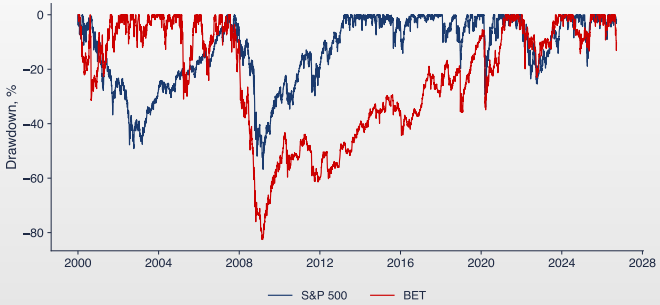
Risk and Return: Interpretation

- **Stock indices cluster** around 15–22% volatility a year
  - returns between 5.0% (WIG20, Warsaw) and 16.6% (Nasdaq 100) a year
- **Crypto assets are far to the right**
  - Bitcoin 66.7%, Ethereum 106.4% volatility a year
- **Long US government bonds (TLT) lost money**: -1.1% a year
  - TLT: an ETF of US Treasury bonds with more than 20 years to maturity; interest rates rose sharply in 2022
- **Question for the room**
  - does the chart prove that higher risk brings higher return?

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  - does the chart prove that higher risk brings higher return?
- **Answer**
  - no: these are averages of one sample; a mean return over 11 years is estimated with a large error, as Chapter 1 shows

Crises and Drawdowns, 2000–2026



- **Drawdown**: the loss from the highest previous price, $DD_t = P_t / \max_{s \le t} P_s - 1$
- $\max_{s \le t} P_s$: the highest price up to day t; $DD_t \le 0$, and $DD_t = 0$ at a new high

SFM_ch0_drawdowns

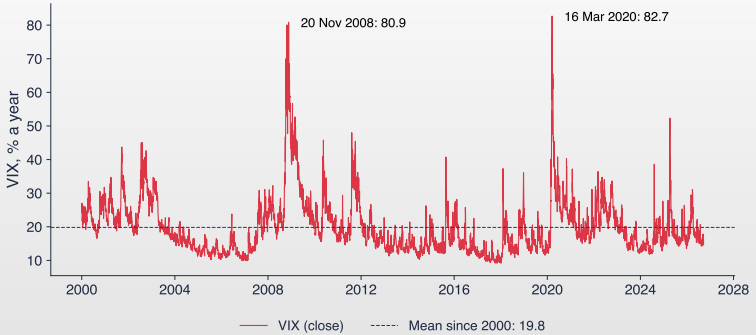
Drawdowns: Interpretation

- **2008 global financial crisis**: the deepest fall of both indices
  - S&P 500: -56.8% at 9 March 2009, from the peak of 9 October 2007
  - BET: -82.5% at 25 February 2009, from the peak of 24 July 2007
- **Later crises**
  - COVID-19, 2020: S&P 500 -33.9%, BET -34.9%
  - rate rises, 2022: S&P 500 -25.4%, BET -22.9%
- **Question for the room**
  - how long did each index need to regain its 2007 peak?

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- **Answer**
  - S&P 500: until 28 March 2013; BET: until 16 March 2021, more than 13 years
  - a deep loss needs a large gain to recover: after -80%, the price must grow five times

The Fear Index: VIX, 2000–2026



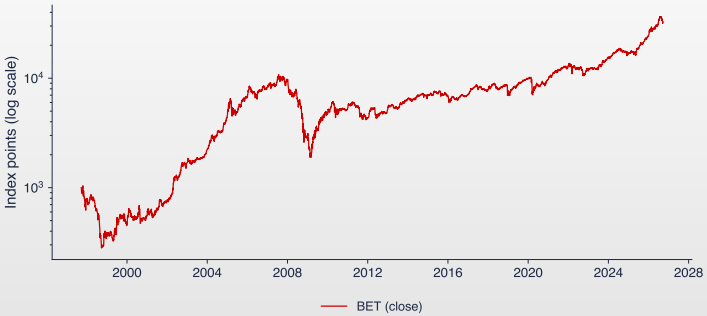
- **VIX**: the volatility of the S&P 500 over the next 30 days expected by option traders, in % a year (Cboe; Whaley (2000))

SFM_ch0_vix

The VIX: Interpretation

- **Usually calm**: mean 19.8, median 17.8 since 2000
  - the mean is above the median: rare spikes pull it up
- **Two spikes above 80**
  - 20 November 2008: 80.9, two months after the bankruptcy of Lehman Brothers
  - 16 March 2020: 82.7, the COVID-19 market panic
- **Volatility comes in waves**
  - high values are followed by high values, and they fade slowly
  - this *volatility clustering* is modelled with GARCH (Generalised AutoRegressive Conditional Heteroskedasticity) in Chapter 9
- On 18 September 2026 the VIX closed at 14.8: a calm market

The Romanian Market: BET since 1997



- **BET**: the reference index of the BVB, launched on 19 September 1997 at 1 000 points (BVB)
- 18 September 2026: 32 758 points, about $\times 32.8$ in 29 years, without dividends

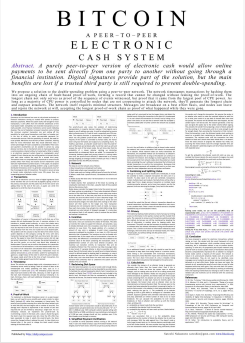
SFM_ch0_bet_history

The Romanian Market: Interpretation

- **Three phases**
  - 1997–1999: a young market loses most of its value
  - 2000–2007: strong growth before the EU (European Union) accession of 2007
  - 2008–2009: a drawdown of -82.5%, then a slow recovery and new highs in 2025–2026
- **Crashes on the BVB can be modelled**
  - log-periodic models fitted to the BET before 2008 (Pele & Mazurencu-Marinescu (2012))
- **Why BET-TR in comparisons**
  - Romanian firms pay large dividends; an index without them understates the investor's gain

Crypto Assets in 2026

- **Bitcoin**: proposed in 2008 (Nakamoto (2008)) as electronic cash without a bank; traded since 2010
- **Closer to mainstream finance**
  - January 2024: the SEC (Securities and Exchange Commission, the US market regulator) approves spot Bitcoin ETFs (SEC (2024))
  - are cryptos becoming an alternative asset class? (Pele, Wesselhöfft, Härdle & Kolossiatis (2023))
- **Still a different statistical world**
  - volatility 66.7% a year since 2015, against 17.7% for the S&P 500
  - trades 24 hours a day, 7 days a week: annualise with 365 days
- Chapter 14 is about crypto assets



BITCOIN

A PEER-TO-PEER

ELECTRONIC

CASH SYSTEM

Abstract. *A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted third party is still required to prevent double-spending.*

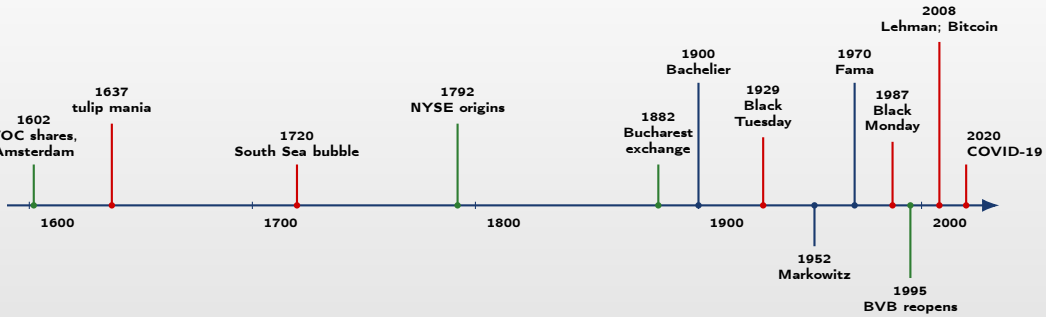
The Bitcoin white paper, 2008

Text: Satoshi Nakamoto (2008); poster: DailyCoinPost (2018); CC0; Wikimedia Commons

Recap: markets in 2026

- Since 2015: stock indices about 15–22% volatility a year, Bitcoin about 66.7%
- Drawdowns can be deep and long: BET -82.5% in 2009, recovered only in 2021
- Volatility comes in waves: the VIX spiked above 80 in 2008 and 2020
- A ranking of markets depends on the window: every average is an estimate

A Timeline of Financial Markets



- Green: exchanges; red: bubbles and crashes; blue: statistical models of prices
- Exchanges, crashes and models grew together: each crisis brought new data and new questions

The First Exchanges: from Bruges to New York



The Amsterdam exchange of Hendrick de Keyser, 1612

Engraving: Claes Jansz. Visscher (1612); public domain; Stadsarchief Amsterdam, Wikimedia Commons

- **Bruges**, 13th–15th centuries: merchants meet in front of the inn of the Van der Beurse family; the name becomes *bourse*, *bursă* (Gelderblom (2013))
- **Amsterdam**, 1602: the Dutch East India Company (VOC) issues shares that anyone can buy and resell
  - continuous trading, forwards and short selling follow within a decade (Gelderblom & Jonker (2004))
- **London**: from 1698, price lists published at Jonathan's Coffee House (Neal (1990))
- **New York**, 17 May 1792: 24 brokers sign the Buttonwood Agreement, origin of the NYSE (New York Stock Exchange) (Sylla, Wright & Cowen (2009))

The First Bubbles



Jan Brueghel the Younger, *Satire on Tulip Mania*, c. 1640

Painting: Jan Brueghel the Younger (c. 1640); public domain; Wikimedia Commons



William Hogarth, *The South Sea Scheme*, 1721

Print: William Hogarth (1721); CC0; National Gallery of Art, Wikimedia Commons

- **Bubble**: a price far above any reasonable value of the asset, followed by a collapse
- **Tulip mania**, Holland, 1636–1637: tulip bulb contracts rise many times, then collapse in February 1637
  - later research: a smaller episode than the legend, with few bankruptcies (Garber (1989); Goldgar (2007))
- **1720**: the Mississippi Company in Paris and the South Sea Company in London rise and fall within months
  - prices rose most for firms tied to a real innovation, Atlantic trade and insurance (Frehen, Goetzmann & Rouwenhorst (2013))
- Recurring ingredients: a new story, cheap credit, then forced selling

Black Tuesday, 1929



Crowd outside the NYSE, 29 October 1929

Photo: U.S. government (1929); public domain; Wikimedia Commons

- **24–29 October 1929**: the New York market crashes; 29 October is **Black Tuesday**
- **The Great Depression**
  - the US contraction that began in August 1929 lasted until March 1933 (NBER)
- **Consequences for statistics**
  - stricter rules on disclosure: firms must publish regular, audited data
  - long drawdowns are not new: the BET needed 13 years after 2008

From the Trading Floor to Screens



The NYSE trading floor, 1963

Photo: Thomas J. O'Halloran (1963); public domain; Library of Congress, Wikimedia Commons



A stock ticker of Thomas Edison

Photo: Jaclyn Nash, Smithsonian (2014); public domain; Wikimedia Commons

- **Stock ticker**, from 1867: prices travel by telegraph, printed on paper strips
  - the first time series of prices available outside the exchange
- **Trading floor**: brokers shout orders in the pit
  - prices recorded on paper by clerks
- **Electronic trading**, from the 1970s–1990s
  - every order and every trade is stored, to the microsecond
  - algorithms send a large share of the orders on major exchanges
- More data made new statistics possible, from daily returns to intraday volatility

The Bucharest Stock Exchange



Brokers at the Stock Exchange Palace, March 1928

Photo: Nicolae Ionescu (1928); public domain; Wikimedia Commons



The same building in 2023

Photo: Chainwit. (2023); CC BY 4.0; Wikimedia Commons

- **1 December 1882**: the Bucharest exchange opens under a decree of King Carol I (BVB)
- **1948**: trading stops under the communist regime; almost 50 years without an exchange
- **20 November 1995**: the first session of the re-established BVB, with 6 listed companies
- **1997**: the BET index starts at 1 000 points
- **2020**: FTSE Russell, a global index provider, upgrades Romania to Secondary Emerging market
- **July 2023**: Hidroelectrica, the largest IPO (initial public offering) in BVB history

Statistics Enters Finance (1/2): Random Walks and Wild Prices



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons

- **1900**: in his doctoral thesis, Bachelier models Paris bond prices as a **random walk** (Bachelier (1900))
- price changes: independent, with the Normal distribution



Benoit Mandelbrot (1924–2010)

Photo: Rama (2007); CC BY-SA 2.0 fr; Wikimedia Commons

- **1963**: Mandelbrot shows that cotton price changes have far heavier tails than the Normal distribution (Mandelbrot (1963))
- he proposes the α-stable distributions of Chapter 3

Statistics Enters Finance (2/2): Risk, Efficiency, Volatility

- **1952**: Markowitz measures risk by the variance of returns and builds diversified portfolios (Markowitz (1952))
  - risk becomes a number that can be estimated
- **1970**: Fama defines the **efficient market**: prices reflect all available information (Fama (1970))
  - testable with statistics: are returns predictable? (Chapter 7)
- **1973**: Black and Scholes price options, with volatility as the key input (Black & Scholes (1973))
- **1982**: Engle models volatility that changes over time: ARCH (AutoRegressive Conditional Heteroskedasticity) (Engle (1982))
  - the start of Chapters 8–10



Eugene Fama at the Nobel Prize ceremony, 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

Modern Crises: 1987, 2008, 2020



Lehman Brothers headquarters, New York, 15 September 2008

Photo: Robert Scoble (2008); CC BY 2.0; Wikimedia Commons



An empty Financial District, New York, March 2020

Photo: Billie Grace Ward (2020); CC BY 2.0; Wikimedia Commons

- **19 October 1987, Black Monday**: the Dow Jones falls 22.6% in one day; large falls in every major market (Roll (1988))
- **15 September 2008**: Lehman Brothers goes bankrupt; the global financial crisis
  - risk models that assumed Normal returns underestimated the losses
- **March 2020**: the COVID-19 crash
  - the S&P 500 log return is -12.8% on 16 March 2020, its worst day since 2000
- Lesson for this course: extreme days are part of the data, not errors to be removed

Recap: a short history

- Exchanges since Bruges (13th–15th centuries), share trading since 1602 (Amsterdam); bubbles and crashes almost from the start
- The Bucharest exchange: 1882–1948, then again from 1995; the BET since 1997
- Models: random walk (1900), portfolio risk (1952), heavy tails (1963), efficiency (1970), changing volatility (1982)
- Each crisis showed that extreme days are more frequent than the Normal distribution predicts

Simple and Log Returns

- P_t: the price at the end of day t (for ETFs and shares, the price adjusted for dividends and splits); P_{t-1}: the price of the previous day
- **Simple return**: $R_t = \dfrac{P_t}{P_{t-1}} - 1$
  - the percentage gain of one day: what the investor earns
- **Log return**: $r_t = \ln \dfrac{P_t}{P_{t-1}} = \ln(1 + R_t)$
  - ln: the natural logarithm; r_t is the continuously compounded return of the day
  - almost equal to R_t for small moves; smaller for large ones
- **Worked example**: $P_0 = 100$, $P_1 = 110$, $P_2 = 99$
  - $R_1 = +10\%$, $r_1 = \ln 1.1 = +9.53\%$
  - $R_2 = -10\%$, $r_2 = \ln 0.9 = -10.54\%$

Which Returns Add Up?

- **Questions for the room**: same example, $100 \to 110 \to 99$
  - what is the sum of the two simple returns?
  - did the investor gain, lose or break even over the two days?

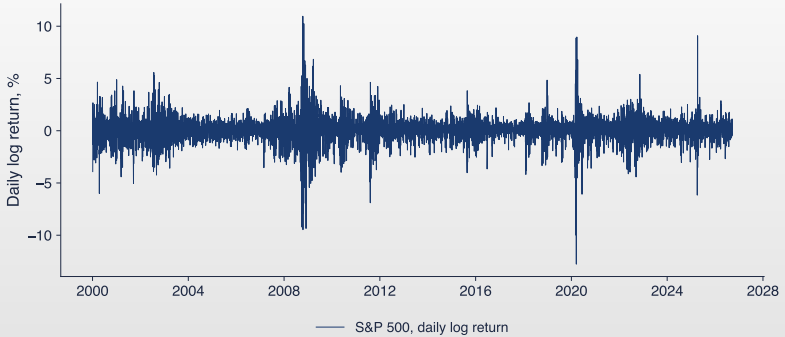
Which Returns Add Up?

- **Questions for the room**: same example, $100 \to 110 \to 99$
  - what is the sum of the two simple returns?
  - did the investor gain, lose or break even over the two days?
- **Answer**
  - $R_1 + R_2 = 10\% - 10\% = 0\%$, but the investor lost: $99/100 - 1 = -1\%$
  - simple returns **compound**: $(1 + R_1)(1 + R_2) - 1 = 1.1 \times 0.9 - 1 = -1\%$
  - log returns **add**: $r_1 + r_2 = -1.01\% = \ln(99/100)$
- Rule: add log returns over time; compound simple returns (details in Chapter 1)

Mean, Volatility and Annualisation

- From n daily log returns $r_1, \ldots, r_n$
  - **mean**: $\bar{r} = \frac{1}{n} \sum_{t=1}^{n} r_t$
  - **volatility**: the standard deviation $s = \sqrt{\frac{1}{n-1} \sum_{t=1}^{n} (r_t - \bar{r})^2}$
  - s: the typical distance of a daily return from its mean; dividing by $n-1$ instead of n corrects for $\bar{r}$ being estimated from the same data
- **Annualisation** with A observations per year
  - mean: $A\bar{r}$, because log returns add up; volatility: $\sqrt{A}\, s$, because the variances of independent days add up
  - $A \approx 252$ for exchanges, $A = 365$ for Bitcoin
- **Worked example**
  - a stock index with daily volatility 1%: $\sqrt{252} \times 1\% = 15.9\%$ a year
  - Bitcoin with daily volatility 3%: $\sqrt{365} \times 3\% = 57.3\%$ (with 252 by mistake: 47.6%)

S&P 500 Daily Returns, 2000–2026



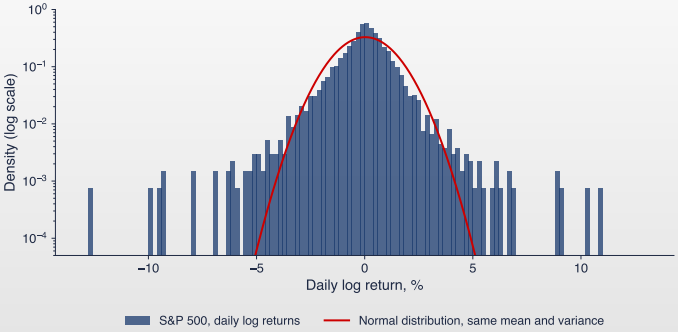
- 6 714 daily log returns, from 4 January 2000 to 18 September 2026

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Daily Returns: Interpretation

- **Small on average, large in crises**
  - mean 0.025% a day, standard deviation 1.21% a day (19.2% a year)
  - worst day: −12.8% on 16 March 2020; best day: +11.0% on 13 October 2008
- **Volatility clustering**
  - large moves come in groups: 2002, 2008–2009, 2020, 2022, April 2025
  - the best and the worst days are close in time
- **The mean is hard to see**
  - the daily mean is about 49 times smaller than the daily standard deviation
  - estimating expected returns needs long samples; estimating volatility is much easier

Returns versus the Normal Distribution



- Histogram of the S&P 500 daily log returns since 2000; red: the Normal density with the same mean and variance; vertical log scale

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Heavy Tails: Interpretation

- **More extreme days than the Normal distribution allows**
  - days with a move larger than 4 standard deviations: 46 observed
  - expected under the Normal distribution with the same mean and variance: 0.43
  - the worst day, 16 March 2020, lies -10.5 standard deviations from the mean
- **Kurtosis** $K = \frac{1}{n}\sum_{t=1}^{n}(r_t - \bar{r})^4/s^4$: the mean fourth power of the standardised returns
  - $(r_t - \bar{r})/s$: distance from the mean in standard deviations; the fourth power lets the extreme days dominate K
  - Normal distribution: $K = 3$; S&P 500 since 2000: 13.7; $K > 3$ means heavier tails than the Normal distribution
- **Also more quiet days**: the histogram is taller in the centre
  - risk measured with the Normal distribution is too low in the tails (Chapters 2, 5 and 10)

Drawdown, Step by Step

t	P_t	peak M_t	DD_t
0	100	100	0%
1	120	120	0%
2	90	120	−25%
3	110	120	−8.3%
4	130	130	0%
5	104	130	−20%

- **Running peak**: $M_t = \max_{s \le t} P_s$, the highest price over the days $s = 0, \ldots, t$
- **Drawdown** (defined with the crises chart): $DD_t = P_t / M_t - 1$
- **Maximum drawdown** (MDD): the most negative DD_t; here −25%, from 120 to 90
- **Why investors watch it**
  - it is the loss of someone who bought at the worst moment
  - it ignores how long the recovery takes: report the duration too

Recap: from prices to returns

- $R_t = P_t/P_{t-1} - 1$ compounds; $r_t = \ln(P_t/P_{t-1})$ adds up over time
- Annualise the mean with A and the volatility with $\sqrt{A}$; $A = 365$ for Bitcoin
- Daily returns: volatility clustering and heavy tails (kurtosis 13.7 against 3)
- Drawdown: the loss from the previous peak; the maximum drawdown summarises the worst episode

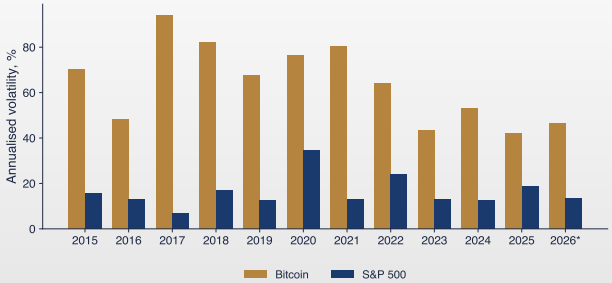
An Open Question

- **Question**: has Bitcoin become less volatile since it entered mainstream finance?
- **Why it matters**
  - a less volatile Bitcoin would be easier to hold in a pension fund or a bank portfolio
  - the spot ETFs of January 2024 opened Bitcoin to investors who buy through ordinary brokerage accounts (SEC (2024))
- **Why it is open**
  - only a few years of data after 2024
  - other explanations exist: a calmer market overall, a larger and more liquid market
  - related evidence: cryptos and traditional assets move closer over time (Pele, Wesselhöfft, Härdle & Kolossiatis (2023))

Possible Uses of AI Tools

- **Literature**: ask for peer-reviewed papers on Bitcoin volatility, with DOIs
  - then open every DOI yourself: AI assistants invent plausible references
- **Hypotheses**: ask for three competing explanations of a fall in volatility
  - each explanation should predict something the others do not
- **Code**: ask for a first draft that computes the volatility per year
  - check the annualisation, the calendar and the dates line by line
- **Critique**: ask the assistant to act as a hostile referee of your result
  - machine learning can generate hypotheses; testing them stays a human job (Ludwig & Mullainathan (2024))

Mini-Case: Volatility per Year



- Annualised volatility of daily log returns in each calendar year; Bitcoin with 365 days, the S&P 500 with its own frequency; 2026*: until 18 September
- Bitcoin: average 69.6% a year in 2015–2023, 47.3% in 2024–2026; in 2025, 2.25 times the S&P 500

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What to Check

- **Fix the comparison before looking at the chart**
  - break date: the ETF approval of January 2024, not the year that looks best
  - trying many dates and keeping the best one produces false discoveries (Harvey, Liu & Zhu (2016))
- **Rule out the rival explanation**
  - if all markets were calmer, the S&P 500 volatility fell too: compare the ratio Bitcoin / S&P 500
- **Check the numbers**
  - 365 days a year for Bitcoin; the same definition of volatility in every year
  - one year is a short sample: a single calm year is not a trend
- **Say what the result cannot show**: a fall after 2024 does not prove that the ETFs caused it

Project Idea

- **Question**: is Bitcoin's volatility, relative to the S&P 500, lower after January 2024 than before?
- **Steps**
  - replicate the chart of the mini-case from the course data
  - write down the hypothesis, the break date and the test before computing
  - compare the variances of daily returns before and after with a test for equal variances (Levene test)
  - repeat with Ethereum as a second crypto asset
- **Report**
  - the two volatilities, the ratio, the test result and one limitation
  - every AI prompt that shaped a decision, in `AI_USE.md`

Check Yourself, and Next: Chapter 1

- **Check yourself**
  - Why is the sum of simple returns not the total return?
  - How do you annualise a daily volatility of Bitcoin?
  - What does a maximum drawdown of -80% mean for an investor?
  - What must you check before using an AI-generated reference?

- **Chapter 1: data, returns and indicators**
  - Where do the data come from?
  - Which price field should we use?
  - How do returns aggregate over days, months and years?
  - What is the Sharpe ratio?
  - What is the volatility drag?

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